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A Novel Approach for Studying the CO₂–Rock Interactions and Estimating the In-Situ CO₂ Volume Using Multifrequency Dielectric Technique

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Abstract

Carbon dioxide (CO₂) is injected into underground reservoirs for several purposes, such as CO₂ sequestration or enhancing the oil recovery (EOR). Various techniques are implemented to investigate the interaction between the CO₂ and rock surfaces. In this work, we propose the use of a multifrequency dielectric approach to understand CO₂-rock interactions and monitor CO₂ saturation.

This study investigates CO₂-rock interactions using carbonate and sandstone core samples. The cores were characterized by measuring porosity, permeability, and mineral composition. CO₂ flooding experiments were carried out under confining and back pressures of 2500 psi and 1500 psi, respectively. Dielectric measurements were performed on dry and CO₂-saturated samples across a wide frequency range (1 MHz to 3 GHz). Low- frequency measurements (1–100 MHz) captured interfacial polarization and rock-fluid interactions, revealing the reactivity and surface complexity. High-frequency measurements highlighted dipolar polarization, enabling CO₂ saturation estimation. The output of dielectric measurements was used to determine the in-situ CO₂ saturation over time and distance from the injection point.

The carbonate samples have an average porosity of 18% and the average sandstone porosity is 20%. The mineralogical composition of carbonate rocks shows around 97% calcite and a few percentages of Illite and Quartz. The sandstone samples composite of 62% Quartz, 23% Plagioclase, and traces of Illite, Ankerite, Chlorite, and Kaolinite minerals. The dielectric profiles carried out on the CO₂/brine saturated cores exhibited a highly dispersive dielectric behavior, with relative permittivity decreasing from 45 at 1 MHz to 7.7 at 100 MHz. After 10 minutes, the dielectric constant in the carbonate sample decreased by a factor of 3.5 within the 1–10 MHz frequency range, indicating strong temporal changes likely due to fluid

redistribution or chemical interactions. In contrast, the sandstone sample showed almost no change over the same frequency range, reflecting greater dielectric stability and lower reactivity. The change in dielectric constant at low frequency is consistent with the fact that carbonate is more reactive compared to sandstone rocks, hence, more dielectric changes are expected for the carbonate samples. The dielectric results were combined with the rock properties and core flooding outcomes to estimate the in situ volume of CO₂.

This work presents for the first time the use of a multifrequency dielectric technique to monitor the CO₂- rock interaction during CO₂ sequestering in carbonate and sandstone formations. The permittivity and conductivity profiles were utilized to estimate the in situ volume of CO₂ trapped as a function of time and distance. Overall, the work findings contribute to a deeper understanding of CO₂ rock interactions and estimating the in- situ CO₂ volume, potentially leading to more effective strategies for CO₂ sequestration.

Keywords: CO₂-rock interactions, dielectric dispersion, in-situ CO₂ volume, multifrequency dielectric

Introduction

Carbon dioxide (CO₂) is commonly injected into subsurface formations for geological storage and enhanced oil recovery (EOR) applications (Aminu et al., 2017; Saraf and Bera, 2023; Bashir et al., 2024; Davoodi et al., 2024). Understanding the interaction between CO₂ and rock surfaces is crucial for evaluating storage security and displacement efficiency (Shukla et al., 2010; Chen et al., 2023). Subsurface CO₂ storage is considered one of the most viable strategies for reducing greenhouse gas emissions (Bashir et al., 2024). Suitable geological formations include deep saline aquifers, depleted oil and gas fields, and unmineable coal seams (Saraf and Bera, 2023; Davoodi et al., 2024). Among these, sandstone and carbonate formations are the most common targets due to their global abundance and favorable reservoir properties (Arif et al., 2017; Yu et al., 2021).

Sandstone reservoirs, composed mainly of quartz grains with relatively low reactivity, offer predictable pore structure and geochemical stability. Their low mineral reactivity means that the primary CO₂ trapping mechanisms include structural, residual, and solubility trapping (Gholami and Raza, 2022; Helmi et al., 2025). However, heterogeneity in grain size and cementation can impact injectivity and sweep efficiency. On the other hand, carbonate reservoirs are more geochemically reactive due to the presence of minerals such as calcite and dolomite. These rocks can undergo mineral dissolution and precipitation when exposed to acidic CO₂-rich brines, leading to enhanced or reduced porosity and permeability depending on local conditions (Siqueira et al., 2017; Massarweh and Abushaikh, 2024; Helmi et al., 2025). This makes carbonate reservoirs complex but potentially valuable for mineral trapping mechanisms, where injected CO₂ can precipitate as solid carbonates (Matter and Kelemen, 2009; Khandoozi et al., 2023; Mouallem et al., 2023).

CO₂ can be injected into the underground formations in different approaches, such as carbonated water or foam (Bashir et al., 2024). The CO₂ injection as foam can improve the mobility control and sweep efficiency by reducing gas channeling and gravitational override, making it effective in heterogeneous formations (Massarweh and Abushaikh, 2024; Rossen et al., 2024). Foam provides greater stability and uniform saturation, especially in high-permeability zones (Abdelgawad et al., 2022; Adebayo et al., 2024; Wang et al., 2024). In contrast, carbonated water injection (CWI) dissolves CO₂ in water before injection, promoting better miscibility and reducing interfacial tension (Esene et al., 2019; Chen et al., 2024). CWI enables enhanced oil recovery while mitigating CO₂ mobility issues, but may offer limited injectivity in tight formations (Esene et al., 2019; Turkson et al., 2024). Both techniques offer advantages in CO₂ storage and EOR, with foam favoring mobility control and CWI enhancing chemical interaction and storage efficiency. Therefore, CO₂ foam will be used in this work in order to improve the CO₂ stability and distribution within the core samples.

Accurate estimation of CO₂ saturation in porous media is essential for monitoring the performance of geological carbon storage and optimizing enhanced oil recovery (EOR) processes (Massarweh and Abushaikha, 2024). Quantifying CO₂ distribution helps assess storage security, identify leakage risks, and improve the efficiency of injected fluids in subsurface reservoirs. Among the various monitoring techniques available, dielectric permittivity measurements have shown significant promise due to their non-invasive nature and sensitivity to changes in both rock matrix and fluid properties (Josh, 2014; Oshaish et al., 2023). These measurements are particularly advantageous for distinguishing between different fluid phases because of the substantial contrast in dielectric constants among rock minerals, water, and CO₂. For example, water exhibits a high dielectric constant (typically >80 at low frequencies), whereas CO₂ and most rock matrices have much lower values, enabling reliable differentiation and phase quantification (Rusiniak, 1998). This contrast forms the basis for using dielectric methods to estimate CO₂ saturation in situ, even under complex multiphase conditions.

The complex refractive index model (CRIM) is widely employed to interpret dielectric responses in porous media by expressing the effective permittivity as a volumetric average of the dielectric constants of the rock matrix and pore-filling fluids (Brovelli and Cassiani, 2008). This mixing model accounts for the geometric and compositional contributions of each phase and is particularly useful in heterogeneous reservoirs. When the porosity of the rock is known, the CRIM equation can be algebraically rearranged to estimate the saturation of individual fluid components (such as CO₂ and water) based on measured effective permittivity. This capability is especially valuable in monitoring CO₂ injection processes, where phase distribution and saturation need to be tracked accurately in real time. Unlike resistivity-based methods, dielectric techniques are largely independent of the ionic strength of the pore fluid, making them more reliable in high-salinity environments and during foam-based CO₂ injection scenarios (Brovelli and Cassiani, 2008; Chen and Heidari, 2018; Azizoglu and Heidari, 2024). As such, dielectric analysis offers a more robust and less ambiguous approach for in situ saturation estimation across a wide range of geological and operational conditions.

This study applies a multifrequency dielectric technique to carbonate and sandstone core samples saturated with CO₂ foam to estimate in situ saturation and monitor rock–fluid interactions. CO₂ foam enhances sweep efficiency and stability, while deionized water with SDS surfactant reduces complications from fluid conductivity. Multifrequency dielectric measurements (1–1000 MHz) were applied using an open-ended coaxial probe to capture both interfacial and bulk polarization effects. Although previous studies explored dielectric responses under gas or brine conditions, little attention has been given to foam injection in reactive and inert rocks. This study addresses that gap by using the CRIM model with explicit porosity to interpret permittivity and quantify CO₂ volume. Low-frequency data provide insights into foam stability, CO₂ movement, and mineral reactivity, while the high-frequency range can be used to estimate the CO₂ saturation. The presented approach can offer a non-destructive and real-time method for evaluating CO₂ behavior in porous media.

Materials and Methods

Materials

In this study, carbonate and sandstone core samples with diameters of 1.5 inches and lengths of 3 inches were used. Routine core analysis was performed to characterize the petrophysical properties of the samples, including porosity and permeability. The measured porosity was 18.5% for the carbonate core and 19.5% for the sandstone core. The carbonate samples exhibit an average permeability of approximately 12 millidarcies (mD), while the sandstone cores demonstrate higher permeability values, averaging around 27 mD. The mineralogical composition of the core samples was determined using X-ray diffraction (XRD) analysis. The carbonate rock consisted mainly of calcite (~97%), with minor amounts of illite and quartz. The sandstone

sample was composed of quartz (62%) and plagioclase (23%), along with trace amounts of illite, ankerite, chlorite, and kaolinite. These compositional differences influence the dielectric behavior for the rock matrix as well as the fluid–rock interactions observed during the experiments.

In addition, deionized water was used as the bulk fluid to minimize the complexity introduced by salinity and to focus on the intrinsic dielectric response of the rock–fluid system. CO₂ foam was generated by injecting CO₂ gas into a surfactant-treated aqueous solution, allowing for stable foam formation. The use of foam helped reduce gas segregation and ensured uniform CO₂ distribution during core saturation, making it suitable for reliable dielectric measurements.

Experimental Procedure

The experimental work includes mainly coreflooding with CO₂ foam and dielectric measurements to evaluate fluid saturation and rock–fluid interactions. The cores were fully saturated with CO₂ foam under controlled conditions to minimize gas segregation and gravity override. CO₂ flooding experiments were carried out under confining and back pressures of 2500 psi and 1500 psi, respectively. A customized stainless-steel cylindrical foam generator was used to produce uniform foam, packed with spherical glass beads to promote a consistent texture. The foam was generated by co-injecting CO₂ gas and deionized water containing 0.3 wt.% sodium dodecyl sulfate (SDS) surfactant, achieving a foam quality of approximately 70%. Dielectric measurements were performed across a frequency range of 1 MHz to 1000 MHz using an open-ended coaxial probe coupled with a Keysight impedance analyzer. Measurements were taken at various locations along the core and at different time intervals to monitor the spatial and temporal evolution of the dielectric response. Relative permittivity and electrical conductivity profiles were obtained for both dry and foam-saturated cores to distinguish between rock matrix and fluid contributions. Finally, the complex refractive index model (CRIM) with explicit porosity inclusion was applied to the dielectric data to estimate in-situ CO₂ saturation. Temporal and spatial variations in the dielectric response were further analyzed to evaluate foam stability, fluid distribution, and potential rock–fluid interactions such as surface reactivity or mineral dissolution.

Results and Discussion

The experimental dielectric measurements and their interpretation for carbonate and sandstone core samples saturated with CO₂ foam are presented and analyzed. Relative permittivity and conductivity profiles were used to estimate in situ CO₂ saturation and assess foam stability within the porous media. The results are further discussed to explore their implications for fluid distribution, rock–fluid interactions, and potential applications in CO₂ storage and monitoring. The discussion begins with the dielectric response of the dry rock matrix and foam-saturated cores, followed by an analysis of CO₂–rock interactions using low-frequency dielectric behavior in both carbonate and sandstone samples. Finally, the dielectric data are used to quantify the in situ CO₂ volume within the core samples.

Analysis of Dielectric Response

This section presents the dielectric permittivity and electrical conductivity responses of carbonate and sandstone cores subjected to CO₂ foam injection. The measurements highlight how fluid saturation and rock type influence frequency-dependent dielectric behavior and can be used to monitor CO₂ distribution. The dielectric properties were measured for carbonate rock samples under both dry and CO₂-saturated conditions. Figure 1 shows the relative permittivity profiles for the dry carbonate matrix and the CO₂-saturated rock, along with the permittivity of the bulk fluid. The dry carbonate matrix exhibits a dielectric constant of approximately 5.7, which is consistent with values reported for Indiana limestone formations. This value remains relatively stable across the measured frequency range of 1 to 1000 MHz, with only minor variations between 5.7 and 5.5. The bulk fluid used in this study is deionized water, which has a dielectric

constant of approximately 81. Deionized water was selected to avoid the interpretational complexity introduced by high-salinity brines, thereby facilitating more reliable estimation of CO₂ saturation. Moreover, the relative permittivity profile of the rock saturated with CO₂ foam shows a decreasing trend with increasing frequency, which is attributed to reduced surface polarization at higher frequencies. In this experiment, CO₂ was injected into the carbonate rock as a foam to minimize gas segregation and gravity override. The use of foam also enhances fluid stability and promotes more effective pore-space trapping, which reduces uncertainty in quantifying the CO₂ volume. At low frequencies (below 10 MHz), some fluctuations in the relative permittivity were observed, likely indicating interactions between the injected fluid and the carbonate matrix. While at higher frequencies, the permittivity values stabilize and remain higher than those of the dry matrix. These high-frequency measurements are particularly useful for estimating in situ CO₂ saturation, as will be discussed in the following sections.

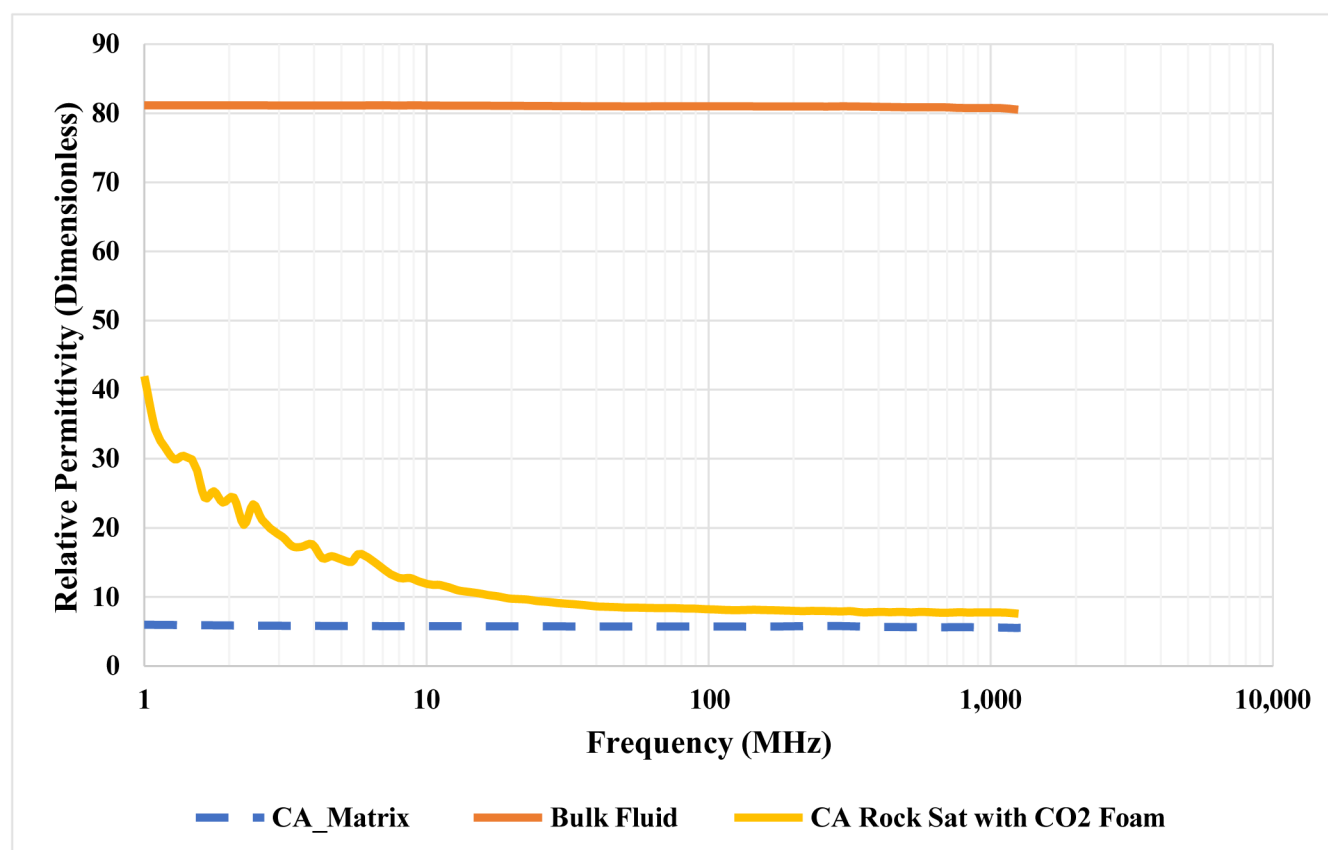


Figure 1—Permittivity profiles as a function of frequency for dry carbonate matrix, CO₂-saturated rock, and bulk fluid.

Moreover, conductivity profiles were measured for the dry carbonate matrix, bulk fluid, and the carbonate core saturated with CO₂ foam, as illustrated in Figure 2. As expected, the bulk fluid (mainly deionized water) exhibits extremely high electrical conductivity, while both the dry rock matrix and the CO₂-saturated core display significantly lower values. This is consistent with the inherently non-conductive nature of carbonate rocks, which limits electrical current flow. Moreover, saturating the carbonate core with CO₂ foam resulted in approximately a twofold increase in conductivity compared to the dry matrix. However, it is important to note that even stabilized CO₂ foam possesses relatively low conductivity. As a result, the overall conductivity of the saturated core remains modest and lower compared to the conductivity of the bulk fluid.

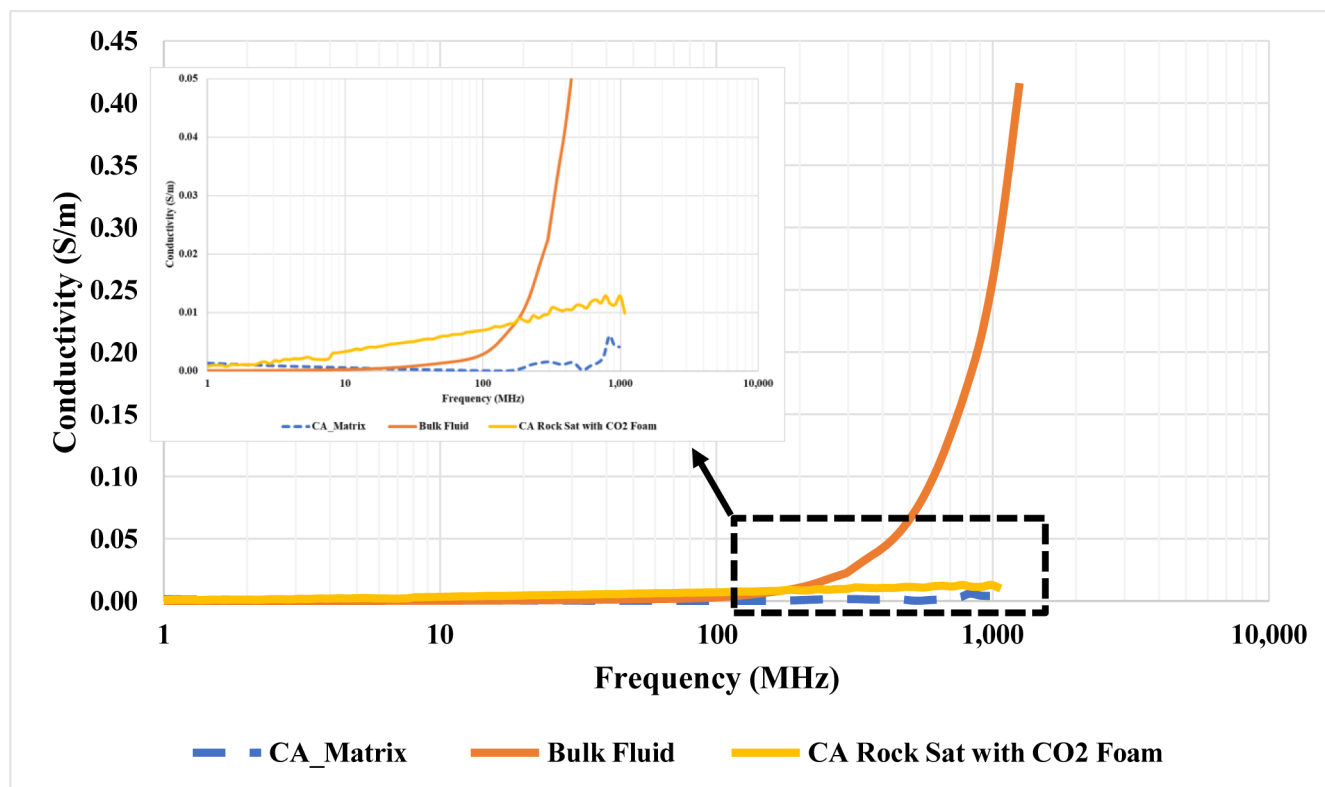


Figure 2—Conductivity profiles as a function of frequency for dry carbonate matrix, CO₂-saturated rock, and bulk fluid

Despite the low conductivity, the measured profile can provide valuable insights into the continuity and distribution of foam lamellae within the pore space, as these structures can facilitate limited current flow. Additionally, variations in conductivity can be used to indicate changes in the rock's internal structure. For example, increases in conductivity may reveal the development of conductivity paths such as vugs or microfractures caused by the carbonate dissolution induced by the injected foam.

In addition, the impact of CO₂ injection into a sandstone formation was investigated by measuring the dielectric properties of both the dry sandstone matrix and the core saturated with CO₂ foam, as shown in Figure 3. The dry sandstone matrix exhibited a relative permittivity ranging from 4.0 to 4.4 across the frequency range of 1 to 1000 MHz. Upon saturation with CO₂ foam, a significant increase in permittivity was observed at low frequencies. Specifically, a dielectric constant of approximately 45 was recorded at 1 MHz, decreasing to 7.7 at 100 MHz. This elevated permittivity at low frequencies is attributed to interfacial polarization caused by the presence of fluids within the pore space. At higher frequencies, only ionic polarization remains active, resulting in lower dielectric values. Moreover, the conductivity profiles were obtained for the dry sandstone matrix and the CO₂ foam-saturated core, as illustrated in Figure 4. Similar to the carbonate samples, the dry sandstone exhibited near-zero conductivity. Saturating the sample with CO₂ foam led to a noticeable increase in conductivity; however, the values remained substantially lower than those of the bulk fluid, consistent with the low conductivity of stabilized foam. It is important to note that the dielectric measurements reported here represent average values along the entire core length. Comparable values were recorded at both the inlet and outlet of the core, indicating good foam stability throughout the sample. In cases where foam stability is poor, gas and fluid segregation can occur, resulting in large variations in dielectric properties along the core. Such variations could serve as effective indicators for evaluating foam stability and spatial distribution. Ultimately, monitoring the permittivity and conductivity over time and distance can help in tracking the CO₂ movement within the formation and assessing the effectiveness of foam-based injection strategies.

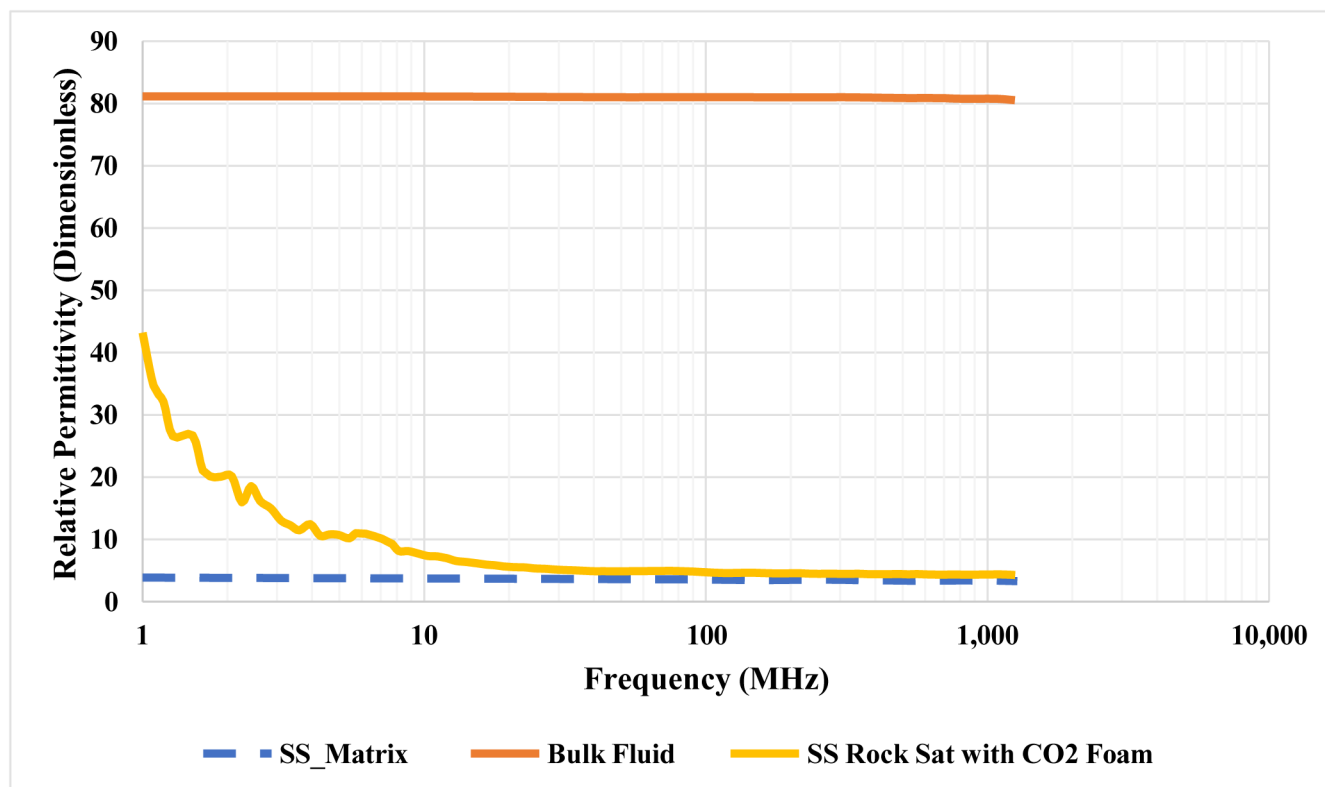


Figure 3—Permittivity profiles as a function of frequency for dry sandstone matrix, CO₂-saturated rock, and bulk fluid.

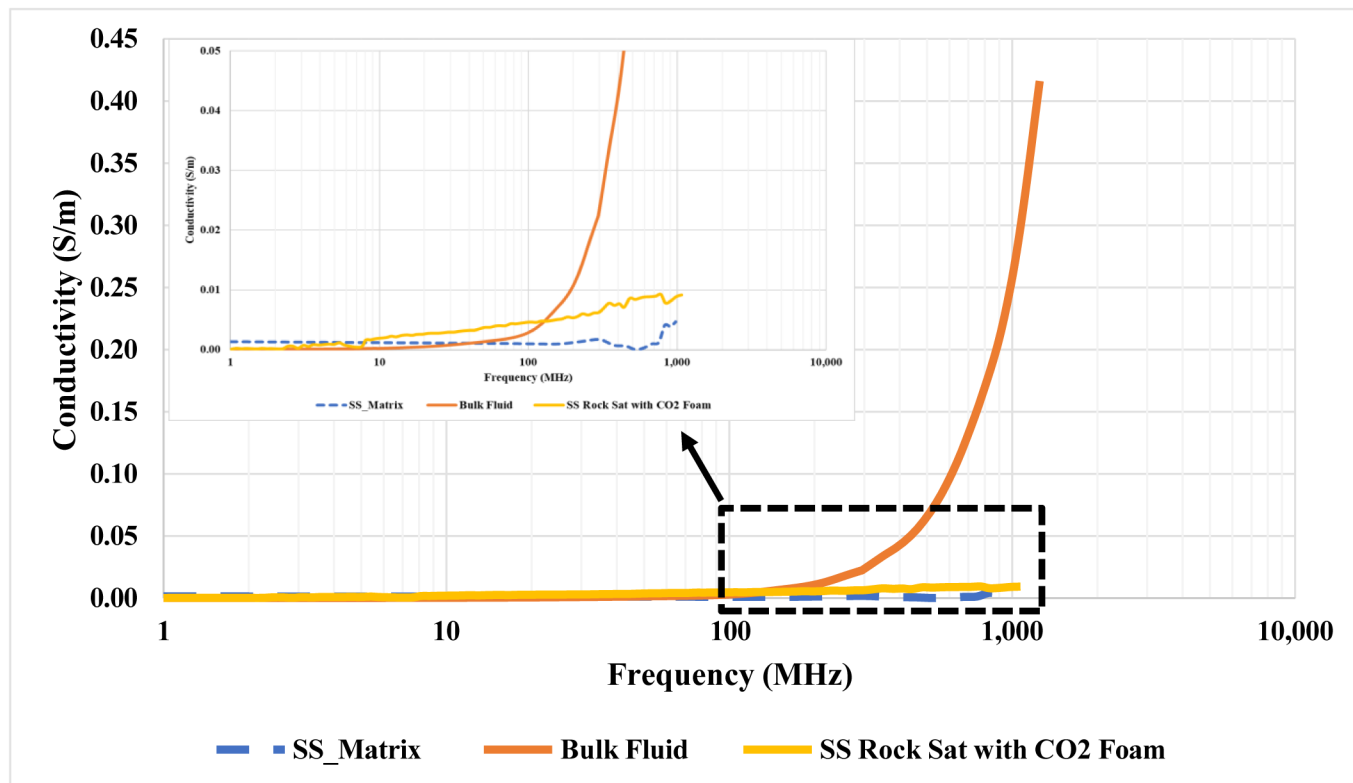


Figure 4—Conductivity profiles as a function of frequency for dry sandstone matrix, CO₂-saturated rock, and bulk fluid.

Monitoring CO₂–Rock Interactions by Low-Frequency Dielectric

The interaction between the injected CO₂ foam and the rock can be effectively studied by analyzing interfacial polarization effects, which are most prominent in the low-frequency range. Figure 5 shows the relative permittivity profiles measured at different core locations and time intervals for a carbonate rock saturated with CO₂ foam. Upon injection, the permittivity at both the inlet and outlet increased from the baseline matrix value of 5.5 to approximately 45, indicating successful foam propagation throughout the core. The overlapping permittivity profiles at the inlet and outlet suggest good foam stability and uniform distribution from the injection point to the production end. However, a noticeable reduction in relative permittivity was observed after ten minutes from stopping the foam injection, with more reductions at the core ends. This decline may be attributed to CO₂ accumulation due to its higher mobility compared to the liquid phase, resulting in rapid gas redistribution. Additionally, changes at the foam–rock interface or reactions between the carbonate matrix and the low-pH injected foam could contribute to the decline in dielectric constant. Such reactions may release CO₂, further influencing the overall permittivity. At high frequencies, the permittivity remained stable, indicating no significant changes in the total gas or liquid volumes. In contrast, the low-frequency response revealed fluctuations, likely due to changes in gas-fluid distribution and the reactive nature of the carbonate surface. These fluctuations reflect interfacial polarization dynamics and possible mineral-fluid interactions. To gain deeper insight into the CO₂–rock interaction and any reaction byproducts, more analyses such as ICP (Inductively Coupled Plasma) spectroscopy are recommended.

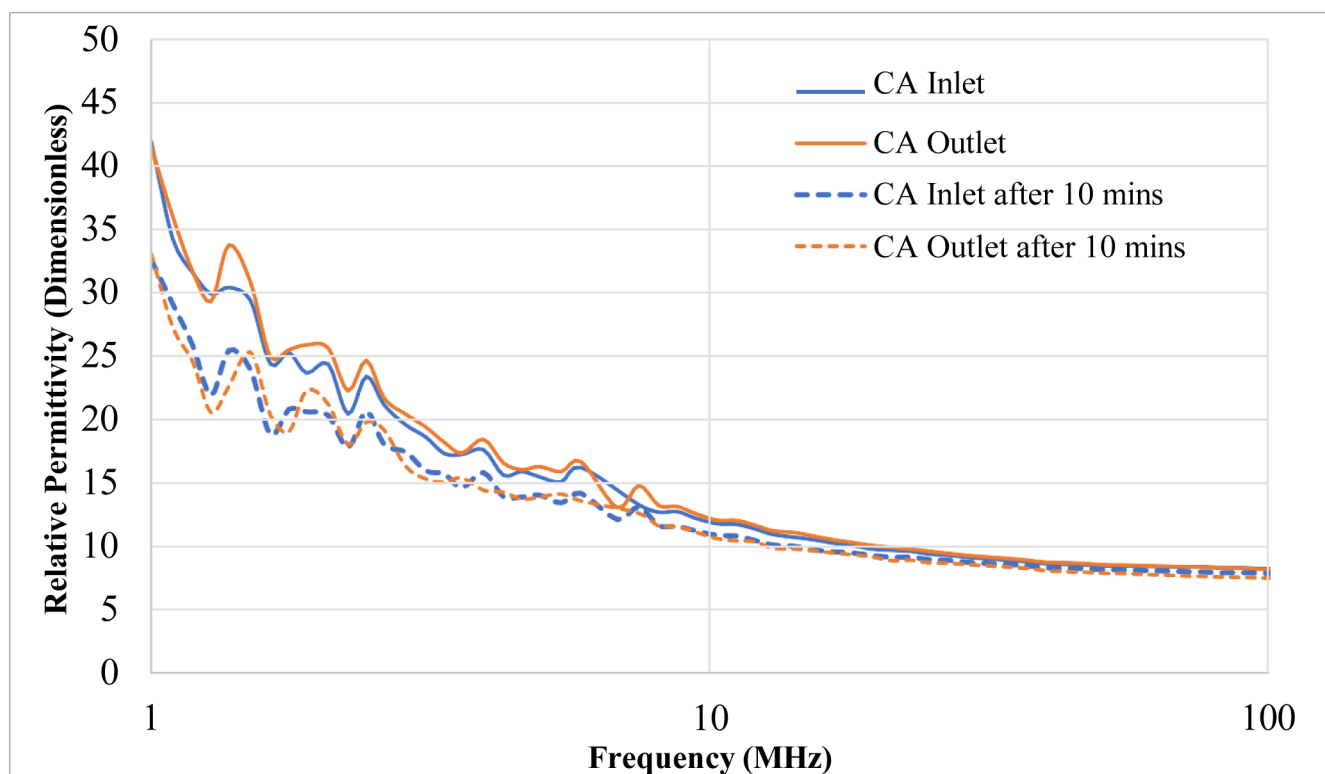


Figure 5—Relative permittivity profiles as a function of frequency for carbonate core saturated with CO₂ foam at different locations and times.

Moreover, dielectric responses in the sandstone core were measured at both the inlet and outlet at two different time intervals, initially and ten minutes after foam injection stopped. The results are shown in Figure 6. The relative permittivity profiles remained nearly unchanged across both locations and times, suggesting minimal interaction between the injected CO₂ foam and the sandstone matrix. A comparison

of the dielectric responses in carbonate and sandstone cores reveals that variations in the low-frequency range are primarily linked to the reactivity between the rock matrix and the injected fluids. Carbonate rocks are more chemically reactive than sandstone; hence, they exhibited greater fluctuations and reductions in permittivity. These fluctuations can indicate the changes in interfacial polarization and modifications to the internal rock surface. Meanwhile, the observed reductions in dielectric constant may be attributed to the generation of additional CO_2 through acid-driven reactions between the carbonate matrix and the injected low-pH foam.

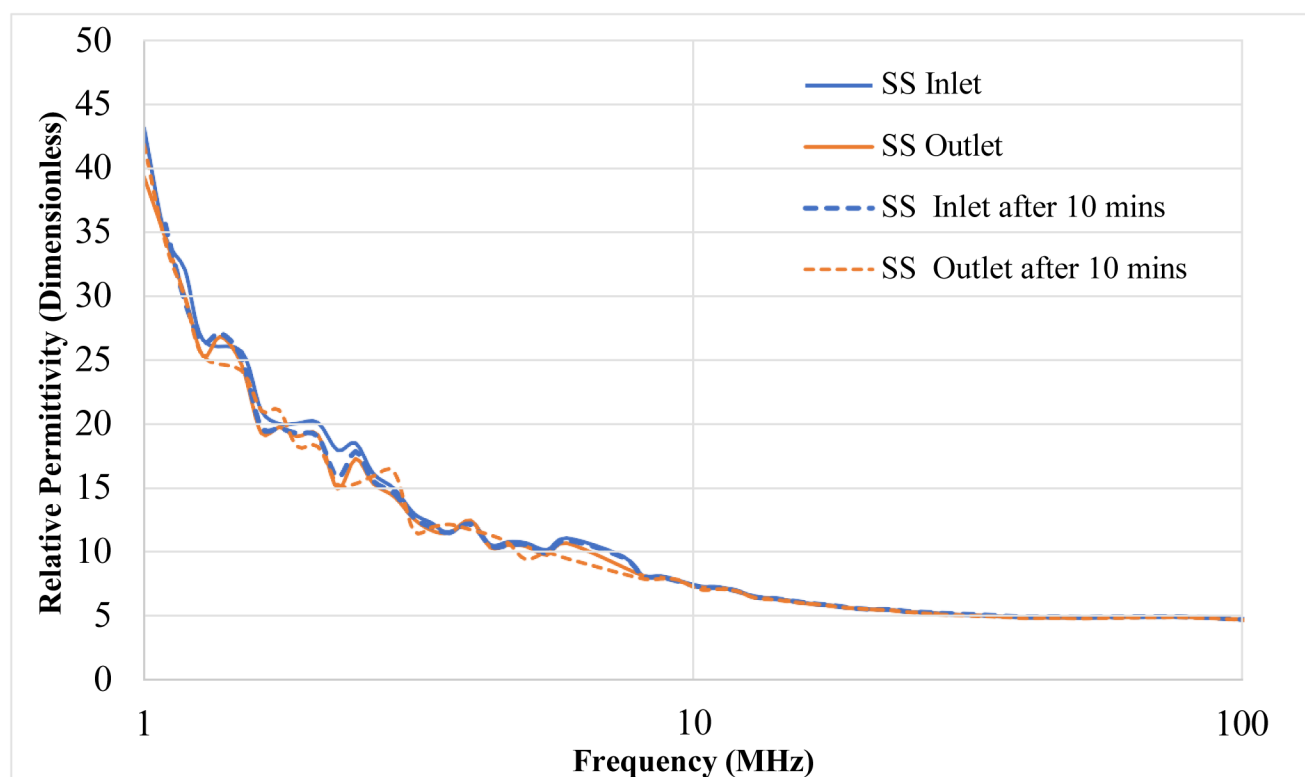


Figure 6—Relative permittivity profiles as a function of frequency for sandstone core saturated with CO_2 foam at different locations and times.

Overall, the dielectric measurements effectively captured the fluid–rock interactions during CO_2 foam injection in both carbonate and sandstone cores. At the low frequency range, significant variations in the relative permittivity were observed in carbonate rocks due to their higher reactivity and interfacial polarization effects. In contrast, the sandstone core showed minimal dielectric changes, indicating lower chemical interaction with the injected foam. Monitoring the dielectric response over an extended period can help identify the point at which chemical reactions reach completion/equilibrium.

CO_2 Volume Estimation Using Dielectric Response

Estimating the CO_2 saturation in porous rocks is critical for evaluating carbon storage and enhanced oil recovery processes. Dielectric measurements offer a non-destructive means to assess fluid saturations by leveraging differences in permittivity between the rock matrix and pore fluids. The complex refractive index model (CRIM), modified to explicitly include rock porosity, enables quantification of CO_2 saturation based on effective permittivity measurements. This approach accounts for the volumetric contributions of the rock matrix, water, and CO_2 , allowing estimation of fluid distributions within the pore space. The following calculations demonstrate how CO_2 saturation can be derived from measured permittivity values and how

variations in bulk fluid permittivity can influence the saturation estimations. CRIM model can be expressed as;

$$\sqrt{\varepsilon_{eff}} = (1 - \varnothing)\sqrt{\varepsilon_{matrix}} + \varnothing(S_w\sqrt{\varepsilon_w} + S_{CO_2}\sqrt{\varepsilon_{CO_2}}) \quad (1)$$

Where ε_{eff} is the effective permittivity of the CO₂-saturated rock, $\varnothing$ is the porosity, ε_{matrix} is the permittivity of rock matrix, S_w is the saturation of bulk fluid, S_{CO_2} is the CO₂ saturation, ε_w is the permittivity of bulk fluid, and ε_{CO_2} is the permittivity of CO₂. The rock is fully saturated then ($S_w + S_{CO_2} = 1$). CRIM equation can be rearranged to estimate CO₂ saturation using known values of permittivity, porosity, and fluid properties. The expression for calculating CO₂ saturation is given as follows:

$$S_{CO_2} = \frac{\frac{\sqrt{\varepsilon_{eff}} - (1 - \varnothing)\sqrt{\varepsilon_{matrix}}}{\varnothing} - \sqrt{\varepsilon_w}}{\sqrt{\varepsilon_{CO_2}} - \sqrt{\varepsilon_w}} \quad (2)$$

The CO₂ saturation can be estimated by applying the CRIM model utilizing the core porosity and dielectric responses. For carbonate rock, the estimated in-situ CO₂ saturation is 34.58% on average. This means that slightly less than half of the pore space is occupied by CO₂, with the remainder filled by brine. This value aligns with the measured effective permittivity being higher than that of the rock matrix, but significantly lower than the bulk fluid, indicating a mixed-fluid system. Such a saturation suggests effective CO₂ penetration and good foam stability within the core sample. The corresponding in-situ CO₂ volume, based on the core's geometry and porosity, is approximately 5.56 cm³. On the other hand, the estimated CO₂ saturation for sandstone sample is around 29.77%, resulting in a CO₂ volume of about 5.04 cm³ based on the core porosity and dimensions. The CO₂ has a very low permittivity; hence the presence of CO₂ can significantly reduce the overall dielectric response, particularly at low frequencies where interfacial polarization is more pronounced. Overall, tracking the dielectric response over time can offer a non-destructive approach to monitor changes in CO₂ saturation and evaluate in-situ CO₂ volume, providing valuable insight into fluid distribution and rock–fluid interactions.

Moreover, the ability to estimate CO₂ saturation and distribution using dielectric permittivity measurements at the core scale demonstrates the potential for applying similar techniques in the field through dielectric logging tools. At the reservoir scale, this approach could enable continuous, real-time monitoring of CO₂ movement, foam propagation, and trapping efficiency during injection operations. The observed sensitivity of low-frequency dielectric response to fluid saturation and rock–fluid interactions suggests that dielectric logging can be used not only for saturation estimation but also for detecting CO₂ breakthrough, foam stability, and possible geochemical reactions. Implementing this method in field deployments offers a non-destructive, high-resolution technique to enhance CO₂ storage monitoring, optimize foam-assisted injection schemes, and improve reservoir management strategies in both carbon sequestration and enhanced oil recovery projects.

Conclusions

This study demonstrated that the multifrequency dielectric technique can be used to study CO₂–rock interactions and estimate in situ CO₂ volume. Measurements of permittivity and conductivity across time and frequency effectively captured fluid distribution and interfacial reactions. In this work, the carbonate cores exhibited strong low-frequency dispersion due to interfacial polarization and mineral reactivity, while sandstone cores showed minimal dielectric variation, reflecting lower chemical interaction with the injected CO₂ foam. The dielectric profiles of CO₂/brine-saturated cores revealed highly dispersive behavior, with relative permittivity decreasing from 45 at 1 MHz to 7.7 at 100 MHz. In carbonate samples, a further reduction by a factor of 3.5 within the 1–10 MHz range was observed after 10 minutes, indicating significant

temporal changes likely driven by fluid redistribution or chemical reactions. Moreover, the CRIM model was used to estimate the CO₂ saturation, yielding values of approximately 34.58% for the carbonate core and 29.77% for the sandstone sample. These saturation levels align with the measured dielectric constants, which reflect the mixed-fluid nature of the pore space. Monitoring the dielectric response over extended time intervals also revealed the progression toward chemical equilibrium, particularly in carbonate rocks where reactive dissolution processes were more pronounced. The obtained results highlight the potential of dielectric methods for real-time monitoring of CO₂ injection, particularly in formations where conventional resistivity-based techniques are limited by salinity or heterogeneity. Future work should focus on integrating dielectric measurements with complementary tools such as ICP, XRD, or NMR to better characterize reaction byproducts and mineral alterations. Additionally, applying this approach in heterogeneous cores and upscaling to pilot-scale or field conditions will help validate its practicality for long-term CO₂ storage monitoring.

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REFERENCES

- Abdelgawad, K.Z., Adebayo, A.R., Isah, A. and Muhammed, N.S., 2022. A literature review of strength and stability of foam and their relationship with the absolute permeability of porous media. *Journal of Petroleum Science and Engineering*, **211**, p.110195.
- Adebayo, A.R., Rezk, M.G., AlYousef, Z., Badmus, S.O. and Babu, R.S., 2024. A Study of CO₂ Storage in Bimodal Carbonate Aquifer Rocks: Challenges and Enhancement through Foaming. *SPE Journal*, **29**(11), pp.6312–6331.
- Aminu, M.D., Nabavi, S.A., Rochelle, C.A. and Manovic, V., 2017. A review of developments in carbon dioxide storage. *Applied Energy*, **208**, pp.1389–1419.
- Arif, M., Lebedev, M., Barifcani, A. and Iglauer, S., 2017. CO₂ storage in carbonates: Wettability of calcite. *International Journal of Greenhouse Gas Control*, **62**, pp.113–121.
- Azizoglu, Z. and Heidari, Z., 2024. Influence of Salt Concentration and Type on Dielectric Permittivity of Rocks. *SPE Journal*, **29**(12), pp.6908–6917.
- Bashir, A., Abdulmoniem, M., Gbadamosi, A., Patil, S., Aljawad, M.S., Mahmoud, M. and Kamal, M.S., 2024. Application of Foam for Enhanced CO₂ Geological Storage: A Mini-Review. *Energy & Fuels*, **38**(20), pp.19208–19221.
- Brovelli, A. and Cassiani, G., 2008. Effective permittivity of porous media: A critical analysis of the complex refractive index model. *Geophysical Prospecting*, **56**(5), pp.715–727.
- Chen, H. and Heidari, Z., 2018. Effect of water salinity and water-filled pore volume on high-frequency dielectric measurements in porous media. *SPE Reservoir Evaluation & Engineering*, **21**(01), pp.202–214.
- Chen, K., Zhang, J.R., Xu, S.Y., Yin, M.Z., Zhang, Y., Zhao, Y.C. and Song, Y.C., 2024. Review of carbonated water injection as a promising technology to enhance oil recovery in the petroleum industry: Challenges and prospects. *Petroleum Science*.
- Chen, Y., Saeedi, A. and Xie, Q., 2023. Interfacial interactions of CO₂-brine-rock system in saline aquifers for CO₂ geological storage: A critical review. *International Journal of Coal Geology*, **274**, p.104272.
- Davoodi, S., Al-Shargabi, M., Wood, D.A., Mehrad, M. and Rukavishnikov, V.S., 2024. Carbon dioxide sequestration through enhanced oil recovery: A review of storage mechanisms and technological applications. *Fuel*, **366**, p.131313.
- Esene, C., Rezaei, N., Aborig, A. and Zendehboudi, S., 2019. Comprehensive review of carbonated water injection for enhanced oil recovery. *Fuel*, **237**, pp.1086–1107.
- Gholami, R. and Raza, A., 2022. CO₂ sequestration in sandstone reservoirs: How does reactive flow alter trapping mechanisms?. *Fuel*, **324**, p.124781.
- Helmi, M.E., Mohammed, I., Rezk, M.G., Gbadamosi, A.O., Raza, A. and Mahmoud, M., 2025. Unlocking the Potential of CO₂ Storage in Saline Aquifers: Challenges, Knowledge Gaps, and Future Directions for Large- Scale Storage. *Carbon Capture Science & Technology*, p.100460.
- Josh, M., 2014. Dielectric permittivity: A petrophysical parameter for shales. *Petrophysics*, **55**(04), pp.319–332.
- Khandoozi, S., Hazlett, R. and Fustic, M., 2023. A critical review of CO₂ mineral trapping in sedimentary reservoirs—from theory to application: Pertinent parameters, acceleration methods and evaluation workflow. *Earth-Science Reviews*, **244**, p.104515.

- Massarweh, O. and Abushaikh, A.S., 2024. CO₂ sequestration in subsurface geological formations: A review of trapping mechanisms and monitoring techniques. *Earth-Science Reviews*, p.104793.
- Matter, J.M. and Kelemen, P.B., 2009. Permanent storage of carbon dioxide in geological reservoirs by mineral carbonation. *Nature geoscience*, 2(12), pp.837–841.
- Mouallem, J., Fathy, A., Arif, M. and Mahmoud, M., 2023, March. CO₂ mineral trapping potential of carbonates: a numerical investigation. In *SPE Middle East Oil and Gas Show and Conference* (p. D021S066R001). SPE.
- Oshaish, A., Hassan, A., Mahmoud, M., El-Husseiny, A., Al-Ofi, S. and Al-Yaseri, A., 2023. Evaluating the rock wettability using multi-frequency dielectric measurements: A review on the fundamental concept and experimental approach. *Journal of Petroleum Science and Engineering*, 220, p.111177.
- Rossen, W.R., Farajzadeh, R., Hirasaki, G.J. and Amirmoshiri, M., 2024. Potential and challenges of foam- assisted CO₂ sequestration. *Geoenergy Science and Engineering*, p.212929.
- Rusiniak, L., 1998. Dielectric constant of water in a porous rock medium. *Physics and Chemistry of the Earth*, 23(9-10), pp.1133–1139.
- Saraf, S. and Bera, A., 2023. Carbon dioxide injection for enhanced oil recovery and underground storage to reduce greenhouse gas. In *Green sustainable process for chemical and environmental engineering and science* (pp. 149–176). Elsevier.
- Shukla, R., Ranjith, P., Haque, A. and Choi, X., 2010. A review of studies on CO₂ sequestration and caprock integrity. *Fuel*, 89(10), pp.2651–2664.
- Siqueira, T.A., Iglesias, R.S. and Ketzer, J.M., 2017. Carbon dioxide injection in carbonate reservoirs—a review of CO₂-water-rock interaction studies. *Greenhouse Gases: Science and Technology*, 7(5), pp.802–816.
- Turkson, J.N., Md Yusof, M.A., Fjelde, I., Sokama-Neuyam, Y.A., Darkwah-Owusu, V., Tackie-Otoo, B.N., Adenutsi, C.D., Amoyaw, B., Hyun, L.J. and Kwon, S., 2024. Carbonated water injection for enhanced oil recovery and CO₂ geosequestration in different CO₂ repositories: a review of physicochemical processes and recent advances. *Energy & Fuels*, 38(8), pp.6579–6612.
- Wang, Z., Wang, P., Li, S., Cheng, H. and Zhang, K., 2024. CO₂ foam to enhance geological storage capacity in hydrocarbon reservoirs. *Renewable and Sustainable Energy Reviews*, 199, p.114504.
- Yu, Y., Hanamertani, A.S., Ahmed, S., Jiao, Z., McLaughlin, J.F. and Quillinan, S., 2021, December. Supercritical CO₂-foam screening and performance evaluation for CO₂ storage improvement in sandstone and carbonate formations. In *Abu Dhabi International Petroleum Exhibition and Conference* (p. D031S097R002). SPE.